SQL QUERY & DATABASE OPERATIONS REPORT

Database Management System Log & Output Analysis

Database: babydino	**Target Table:** products	**Environment:** MySQL Workbench 8.0

1. TABLE DEFINITION (DDL)

```
CREATE TABLE products (
    Product_ID VARCHAR(10) PRIMARY KEY,
    Product_Name VARCHAR(100),
    Category VARCHAR(50),
    Price INT,
    Size VARCHAR(20),
    Color VARCHAR(30),
    Stock_Quantity INT
);
```

2. EXECUTED QUERIES & DATA OUTPUT

Query 2.1: Retrieve All Products

```
SELECT * FROM products;
```

Product_ID	Product_Name	Category	Price	Size	Color	Stock_Quantity
P001	Cotton Round Neck T-Shirt	T-Shirt	799	S	Black	45
P002	Cotton Round Neck T-Shirt	T-Shirt	799	M	Black	40
P003	Premium Polo T-Shirt	T-Shirt	999	L	Navy Blue	32
P004	Oversized Graphic Tee	T-Shirt	1099	XL	White	28
P005	Slim Fit Shirt	Shirt	1299	M	Sky Blue	35
P006	Casual Linen Shirt	Shirt	1499	L	Beige	22
P007	Formal White Shirt	Shirt	1399	M	White	30
P008	Denim Jacket	Jacket	2499	L	Blue	18
P009	Hooded Sweatshirt	Hoodie	1899	M	Grey	25

Query 2.2: Filter Products with Low Stock (< 20 units)

```
SELECT * FROM products WHERE Stock_Quantity < 20;
```

Product_ID	Product_Name	Category	Price	Size	Color	Stock_Quantity
P008	Denim Jacket	Jacket	2499	L	Blue	18
P014	Regular Fit Jeans	Bottom Wear	1899	34	Black	17
P017	Running Shoes	Footwear	2999	8	White	15
P018	Sneakers	Footwear	3499	9	Black	12
P025	Travel Backpack	Bags	2199	Free	Black	16
P026	Laptop Backpack	Bags	2799	Free	Grey	13
P027	Duffel Bag	Bags	2499	Free	Navy Blue	11
P028	Gym Bag	Bags	1899	Free	Black	18

Query 2.3: Total Products Grouped by Category

```
SELECT Category, COUNT(*) AS Total_Products
FROM products
GROUP BY Category;
```

Category	Total_Products
T-Shirt	4
Shirt	3
Jacket	1
Hoodie	2
Bottom Wear	4
Shorts	2
Footwear	2
Accessories	8
Bags	4

Query 2.4: Top 5 Highest Priced Products

```
SELECT * FROM products
ORDER BY Price DESC
LIMIT 5;
```

Product_ID	Product_Name	Category	Price	Size	Color	Stock_Quantity
P018	Sneakers	Footwear	3499	9	Black	12
P017	Running Shoes	Footwear	2999	8	White	15
P026	Laptop Backpack	Bags	2799	Free	Grey	13
P008	Denim Jacket	Jacket	2499	L	Blue	18
P027	Duffel Bag	Bags	2499	Free	Navy Blue	11

Query 2.5: Calculate Average Product Price

```
SELECT AVG(Price) AS Average_Price
FROM products;
```

Average_Price
1445.6667